

COMPSCI 752

Course Administration

Big data management

Instructors: Karuna Reddy, Philipp Skavantzios and Sebastian Link

Course Overview in a Nutshell

Principles and Technologies for Big Data

- Need to manage large volumes of diverse data to stay competitive
- Need to master many technologies reliant on new principles to represent, describe and access data

Big data principles

- Prepare you for a career as data scientists, independently of continuous technology changes
- Learn how to model, query, and integrate big data
- MapReduce algorithms such as PageRank illustrate how to efficiently rank billions of data elements

Big data technologies

- What is and what is not possible today?
- Spark: technology framework for preparing and analysing big data
- Blockchain: distributed computing system with high fault tolerance

COMPSCI 752 Lecturer

**Dr
Karuna Reddy**

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Office:

Building: 303

Room: 418

Hours: by appointment



COMPSCI 752 Industry Perspectives

- Advisors will provide an industry perspective for some of the content in 752
- Real-life use cases and tools, with focus on graphs and semantic technologies
- Tell you about skills required to succeed within industry
- We reserve one hour slots for every other week
- Slots will be flexible pending on availability of the advisors
- Slots will be announced within the week prior

Course Content



Part 1: Big data models (Weeks 1-5)

- XML and graph models
- Query languages and design

Part 2: Big data semantics (Week 6-7)

- Ontology and knowledge graphs
- Big data algorithms

Part 3: Big data apps (Week 8-10)

- Map/Reduce
- Streams and Privacy

Part 4: Present & Revise (Week 11-12)

- Watch research presentations
- Revision of material for exam

Skills for your CV (Learning outcomes)

- 1 Apply state-of-the-art in representation formalisms for big data, including XML, RDF, JSON, Graphs, and NoSQL
- 2 Model big data with schema languages, including DTDs, XML Schema, JSON schema, Property graph models, Neo4j, RDF schema
- 3 Access big data with query languages, including XPath, XQuery, Cypher, SPARQL, Spark SQL
- 4 Reason about big data, including the Web Ontology Language OWL, Knowledge Graphs
- 5 Understand how to manage and analyse big data, including techniques for searching and processing such as PageRank, MapReduce, Spark and Blockchain
- 6 Present as team your joint understanding of state-of-the-art knowledge on a topic about big data
- 7 Communicate your understanding about a topic on big data, including fair use of this knowledge

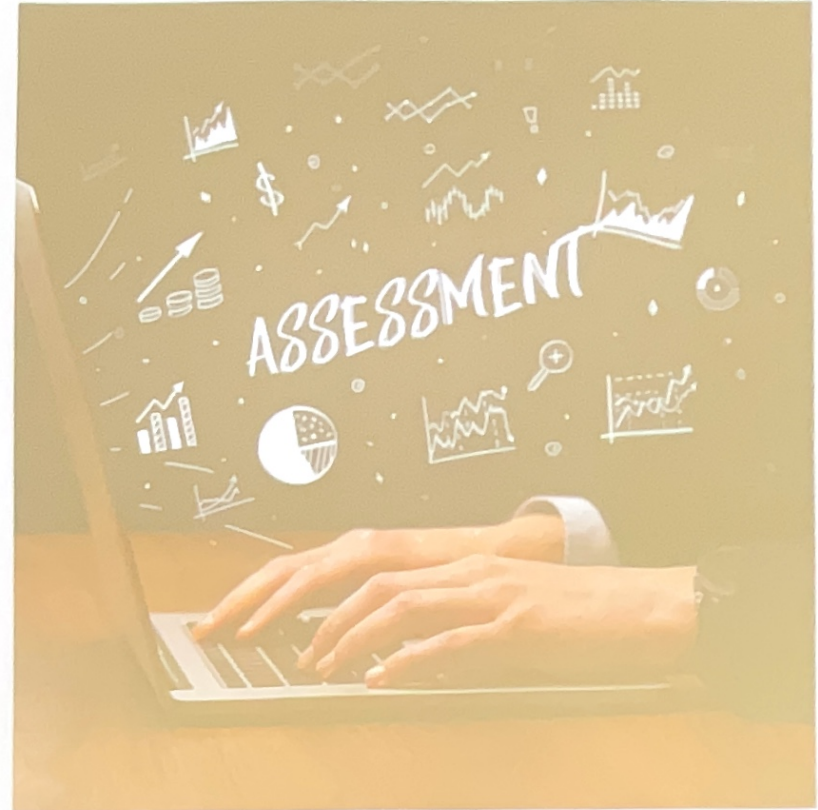
Learning Resources

- All material will be posted on CANVAS
 - ▶ Slides in pdf
 - ▶ Video recordings of lectures
 - ▶ Video recordings of tutorials and solutions to exercises
- Search and study articles and text books
- Revisit material
- Do strategic exercises and check model answers
- Form a study group and discuss with peers
- Think of examples
- Watch presentations
- Program and experiment



Assessment - Overview

- Four assignments (25% in total, 6.25% each)
- One small-team presentation (25%)
- Final exam (50%)



Assessment - Details

Four Assignments

- Short answers, style similar to exam
- Due by Fri, 28 March 2025 (Week 4)
- Due by Fri, 11 April 2025 (Week 6)
- Assignments 3 and 4 tba
- Released a week ahead of due data



Presentation

- Based on topic of research article
- In teams of 3-4
- 16 minutes in total
- Marking guidelines on CANVAS
- Material due by Fri, 23 May 2025 (W10)

Final Exam

- Invigilated, in-person exam
- Two (2) hours
- Open book
- Time tba

Presentations - Topical areas to choose from



T1: Exchange

- 1) XML design
- 2) XML queries
- 3) XML schema
- 4) DTD inference
- 5) JSON data
- 6) JSON schema
- 7) JSON design
- 8) JSON inference

T2: Graphs

- 1) PG-Query
- 2) PG-Keys
- 3) Visualization
- 4) PG-Schema
- 5) Parallelism
- 6) Benchmarks
- 7) Communities
- 8) MapReduce

T3: Semantics

- 1) Linked data
- 2) OWL
- 3) Sparql
- 4) Reverse Sparql
- 5) Schema.org
- 6) Faceted Search
- 7) Completion
- 8) Trust

T4: Knowl Graphs

- 1) Refinement
- 2) Foundations
- 3) Architecture
- 4) Construction 1
- 5) Construction 2
- 6) Semantics
- 7) Querying
- 8) Successes

T5: Profiling

- 1) Overview
- 2) FDs
- 3) Hybrid FD
- 4) Dynamic FD
- 5) Uniques
- 6) Foreign keys
- 7) SQL keys
- 8) MVDs

Presentation - Topical Areas Continued

T6: Spark

- 1) Overview
- 2) Core
- 3) Streaming
- 4) SQL
- 5) Mach. Learning
- 6) Graphs
- 7) Map/Reduce
- 8) Reuse

T7: Blockchain

- 1) Architecture
- 2) Crypto basics
- 3) Contracts
- 4) Permissionless
- 5) Mining
- 6) Permissioned
- 7) Scalability
- 8) Signing

T8: NoSQL

- 1) C-Store
- 2) Aurora
- 3) Flink
- 4) DocumentDB
- 5) Dynamo
- 6) Big table
- 7) In-memory
- 8) Delta Lake

T9: LLM&VecDB

- 1) LLM
- 2) Milvus
- 3) Manu
- 4) PASE
- 5) VBASE
- 6) AnalyticDB-V
- 7) Deep Lake
- 8) Cloud

T10: ML Systems

- 1) Federated Learning
- 2) Snorkel
- 3) Pytorch
- 4) TensorFlow
- 5) Tensors
- 6) Tensor Compiler
- 7) Tensor Query
- 8) WindTunnel

Presentation - Topic Selection

- Scan the articles in CANVAS, and read them a bit (for example their abstract or introduction)
- **Send 3 choices of topics to all lecturers** by Fri, 14 March 2025, eg.:
 - 1 T9.8
 - 2 T2.1
 - 3 T5.5
- List choices from most to least preferred
- Lecturers will compile and publish list of topic assignments
- Find and read relevant literature on your topic
- Formulate a research question and select 5 reference articles for topic by Fri, 04 April 2025
- Send the question and references to these articles to **all** lecturers by that time
 - ▶ *karuna.reddy@auckland.ac.nz*
 - ▶ *philipp.skavantzios@auckland.ac.nz*
 - ▶ *s.link@auckland.ac.nz*

Presentation

- Team consists of 3-4 people
- Develop a professional presentation for the assigned topic jointly with your team partners
- Duration is 16 minutes and should effectively convey the most critical knowledge to your peers
- Each team member should contribute an equal amount of effort and time to the presentation
- All presentations must be recorded and submitted by the due date
- Marking guidelines are available on CANVAS

Academic Integrity

<https://academicintegrity.cs.auckland.ac.nz>



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Timetable

Allocated Lectures/Tutorials/Industry perspective:

- Tue 03:00PM-05:00PM, 303-G23 (MLT1 Lecture Theatre)
- Fri 10:00AM-12:00NOON, 405-460 (7 Grafton Road)

Organization:

- Live lectures and tutorials
- Lecture and tutorial recordings
- Study material by yourself at your own pace every week



Timeline (all 2025 dates)

 Sent to 3 email.

Week 2: Topic choices

due by Friday, 14 March

Week 4: Assignment 1

due by Friday, 28 March

Week 5: Research question & references for presentation

due by Friday, 04 April

Week 6: Assignment 2

due by Friday, 11 April

Break: 14 April - 25 April

Week 7-11: Assignments 3 and 4

due dates tba

Week 10: Team presentation material

due by Friday, 23 May

Big data and its characteristics

Big data is the data asset characterized by such a variety, velocity, veracity, and volume to require specific technology and analytical methods for its transformation into value.

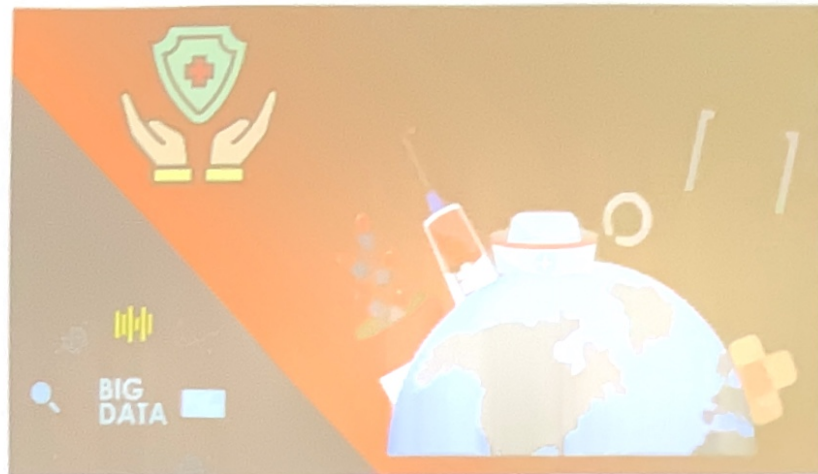
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Top Big Data Use Cases in Healthcare

- **Predictive analytics:** Analyse vast patient data to take early actions
- **Personalised medicine:** Identify targeted treatments from genetic profile and lifestyle
- **Remote patient monitoring:** Find patterns and trends, make consultations and treatment accessible
- **Health data analytics:** Insight from population health patterns to develop interventions and prevent outbreaks
- **Drug discovery and development:** Analyse data to find new drugs, forecast efficacy and safety, and improve clinical trial design
- **Operational efficiency:** Optimize patient scheduling & resource allocation to improve operations, patient care and outcomes



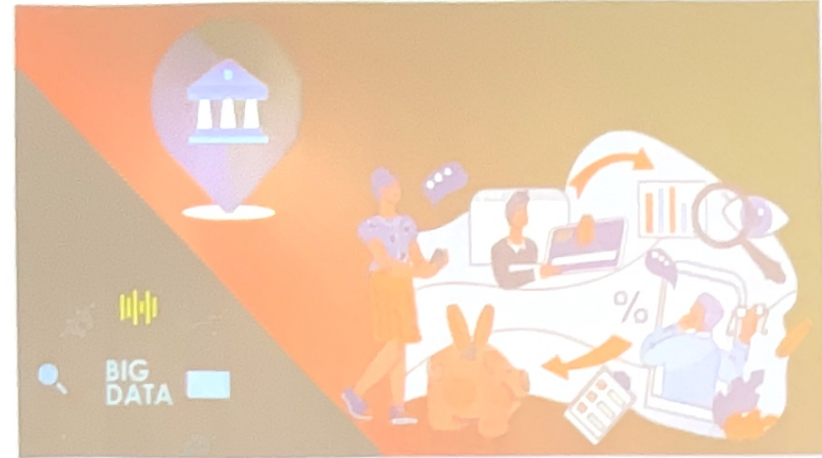
Top Big Data Use Cases in Retail

- **Personalised recommendations:** Analyse preferences & behaviors from browsing history, purchase behavior, social media
- **Inventory optimisation:** Analyse historical and real-time log data on sales, returns, stock levels to optimise inventory levels
- **Price optimisation:** Analyse competitor pricing, historical sales, customer demand & market trends to identify optimal price points
- **Supply chain management:** Analyse log data on logistics, transportations, and inventory levels to optimise supply chain
- **Fraud detection:** Detect and prevent fraud in online transactions, credit card processing and loyalty programs
- **Market trend analysis:** Analyse social media, customer reviews, online forums to understand customer sentiments and trends



Top Big Data Use Cases in **Banking and Financial Services**

- **Fraud detection and prevention:** Spot anomalies in purchasing patterns or frequent alterations to account information
- **Risk management:** Better credit, market & operational risks by analysing market trends, economic indicators & customer behaviour
- **Customer analytics:** Improve experience, personalise services, identify opportunities by analysing customer data
- **Compliance and regulatory reporting:** Analyse accounts & behaviour of customers to monitor compliance and avoid penalties
- **Trading and investment analytics:** Analyse market trends, financial data, portfolios to inform trading & investment decisions
- **Loan management:** Analyse credit risk, borrower eligibility & predict default rates to improve processes & reduce risk of default



Conclusion

- Develop smart solutions by gaining insight from storing, analysing and managing big data
- Companies enhance operations, deliver superior customer service, craft personalised marketing campaigns, and undertake initiatives that ultimately amplify revenue and drive greater profits
- Benefits are not just financial but impact the insight on life as a whole and therefore its quality
- The outcomes of any analytical methods (eg AI) are limited by the quality of data
- Shift from re-active to pro-active behaviour
- Success = *Action* + insight generated from data

Focus of this course

Big data characteristics in focus

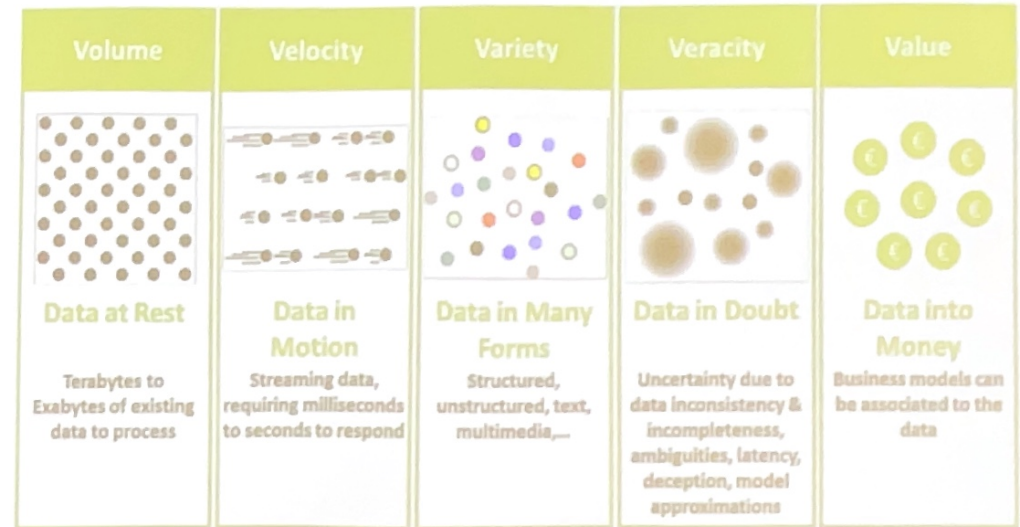
- Primary: Volume & Variety
- Secondary: Velocity & Value

Principles and technologies

- Principles: describe & access big data
- Technologies: manage big data

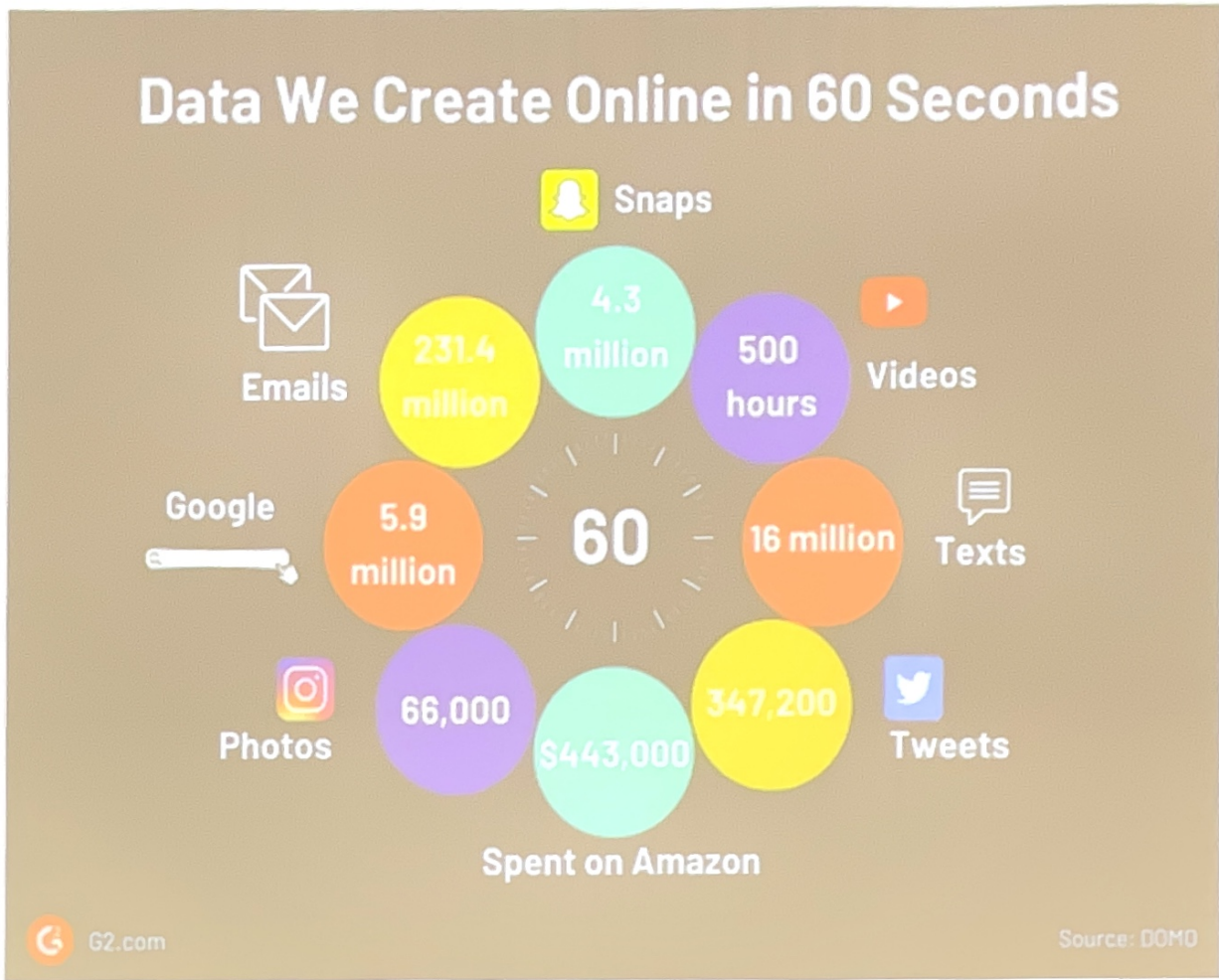
Core course aims

- Effective modeling of big data
- Reasoning about big data
- Big data management and distribution



Integrated by a post of Michael Wraith on 28 November 2012

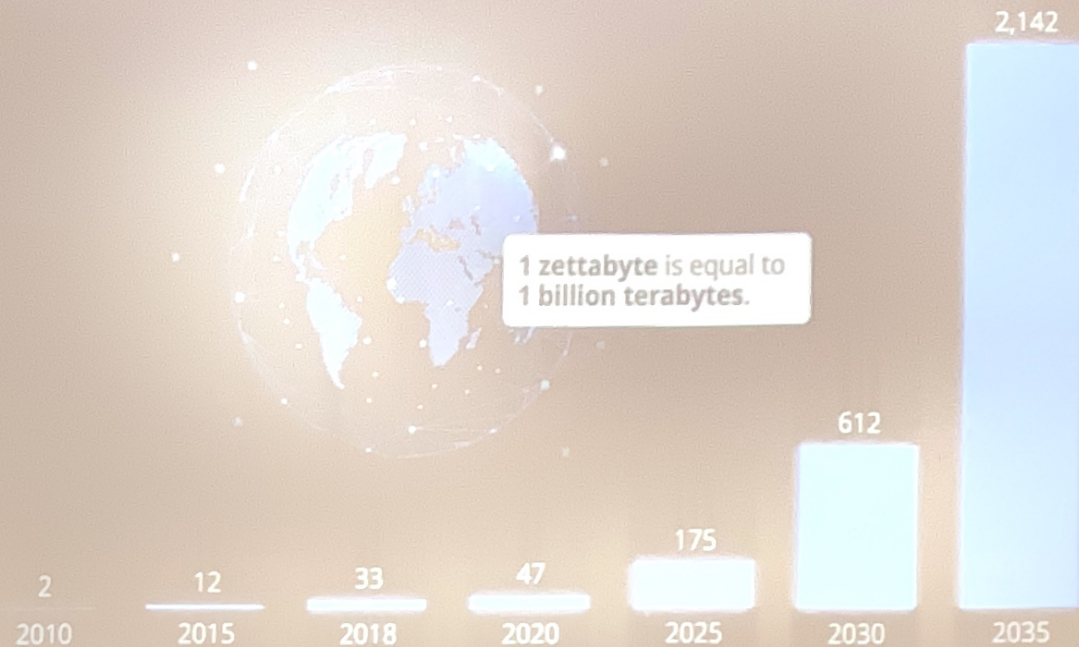
Big Data - So hot, so now, so wow



Big Data: Volume

Global Data Creation is About to Explode

Actual and forecast amount of data created worldwide 2010-2035 (in zettabytes)



@StatistaCharts

Source: Statista Digital Economy Compass 2019

statista

Big Data: Variety

Types of Big Data

Structured

Pre-defined data models like databases

Usually text only

Easy to search and filter

Examples:
Dates, phone numbers, transaction information



Semi-Structured

Both structured & unstructured qualities

Considerably easier to analyze than unstructured data

Examples:
Emails, CSV files, JSON files



Unstructured

No Pre-defined data models

Difficult to search through

Usually stored as different types of files

Examples:
Social media data, audio files, images



Big Data: Veracity 真实性.

Sources of Data Veracity



Statistical biases



Lack of data lineage



Software bugs



Noise



Abnormalities



Information Security



Untrustworthy data sources



Falsification



Uncertainty and ambiguity of data



Duplication of data



Out of date and obsolete data

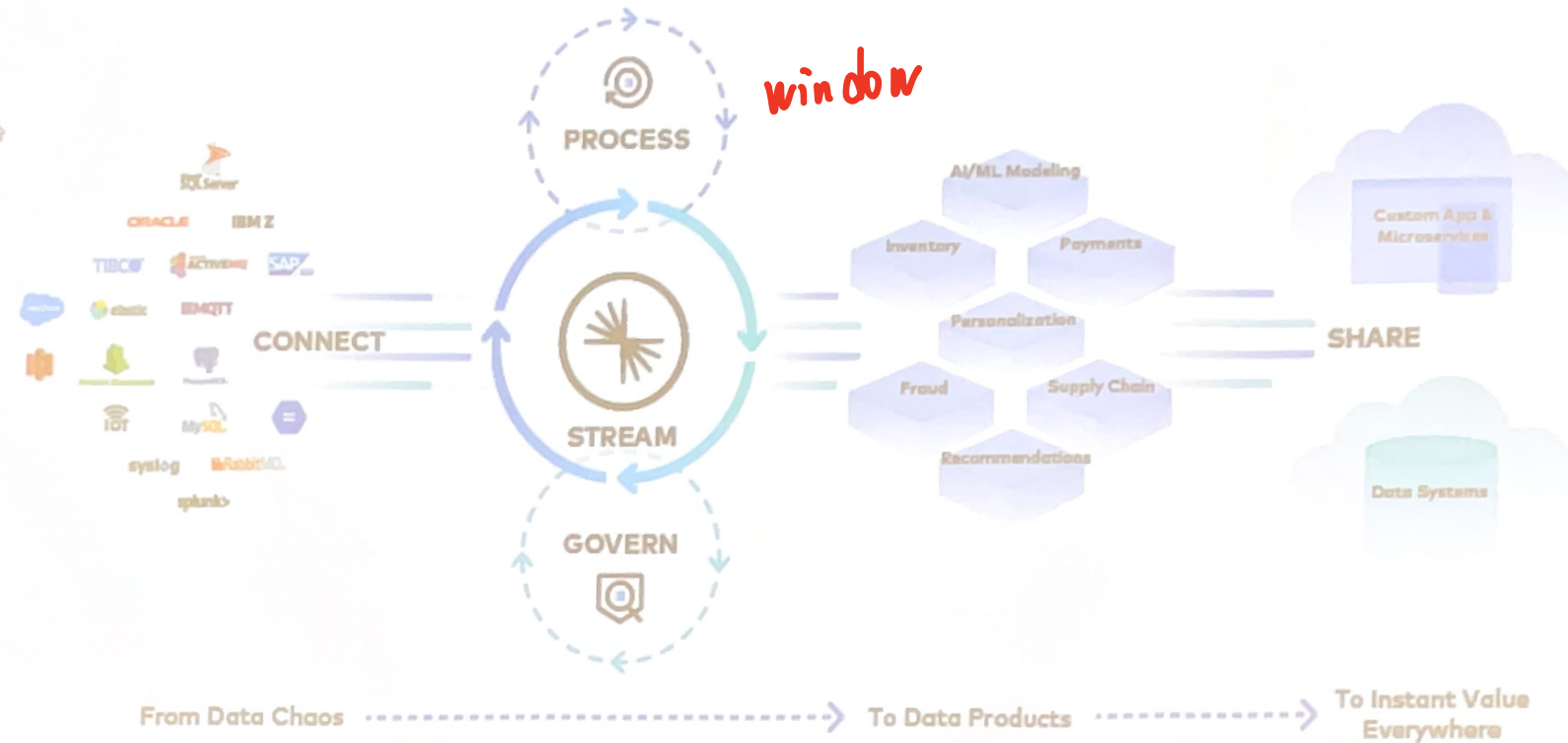


Human error

Big Data: Velocity

A New Paradigm for Data in Motion: Data Streaming

All your data continuously streamed, processed, governed and shared as a product, making it instantly valuable, usable, and trustworthy everywhere.



Big Data: Value

